

TUSHAR KATYAL

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EDUCATION

GGSIPU, Delhi MCA (Software Engineering) · <i>GPA: 9.52</i>	2023 – 2025
P.G.D.A.V. College, Delhi University B.Sc (H) Computer Science · <i>GPA: 9.365</i>	2020 – 2023
RDJK (G) Co-ed Sr. Secondary School, Delhi Class 12th Science · <i>GPA: 93%</i>	2019 – 2020
Pt. Yad Ram Secondary Public School, Delhi Class 10th · <i>GPA: 93.6%</i>	2017 – 2018

WORK EXPERIENCE

TO THE NEW <i>Data Engineer</i>	Noida, India January 2025 – Present
- Designed and implemented cloud-based data pipelines on AWS using S3, EC2, Glue, EMR, Lambda, Athena, and PySpark to optimize big data processing and cloud data warehousing. - Developed scalable ETL pipelines using Apache Airflow to automate complex data ingestion, transformation, and workflow scheduling.	
University of Delhi <i>VCIS Intern</i>	Dean Students' Welfare, Delhi June 2022 – August 2022
Authored the official Culture Council Annual Report presented to the Vice Chancellor and built the department's web application to streamline communications.	

PUBLICATION

Applications of Python in Solving Physical Chemistry Problems (Part 1) <i>Resonance – Journal of Science Education (Indian Academy of Sciences / Springer)</i>	Dec 2025
Co-authored research on integrating Python for computational chemistry, focusing on generating pH-metric and potentiometric titration curves using Matplotlib and NumPy.	
<i>Link: https://rdcu.be/eVVU0</i>	

SKILLS

Programming Languages:	Python, SQL, Java, C++, HTML, CSS, JavaScript
Cloud & Big Data Tools:	PySpark, AWS (S3, Glue, Athena, EC2, Lambda, EMR), Kafka, Airflow, Hadoop, Hive
Database Management:	PostgreSQL, MySQL, MongoDB

PROJECTS

OmniRoute: Smart Logistics Engine <i>PySpark, Kafka, Airflow, AWS (S3, Glue, EC2), PostgreSQL</i>
- Designed and implemented an enterprise-grade Data Lakehouse architecture to ingest and process high-velocity telemetry data from a global fleet of delivery vehicles. - Developed a robust batch pipeline using AWS Glue and PySpark to manage fleet assignment logs with SCD Type 2 logic, enforcing historical traceability and window-function based conflict resolution. - Built a real-time Kafka streaming architecture evaluating GPS and sensor telemetry data against geofences and speed limits to instantly apply algorithmic driver safety strike penalties. - Automated daily fuel consumption auditing and monthly rate-deduction reports using Apache Airflow orchestrating Egress pipelines from S3 Delta layers into a PostgreSQL query-optimized database on EC2.
Nimbus Data Accelerator <i>PySpark, Airflow, Kafka, AWS (S3, EMR, EC2, Athena), Python, SQL</i>
- Developed and orchestrated a robust data validation, profiling, and quality check framework as part of the Nimbus data accelerator pipeline project. - Designed EMR-based PySpark workflows integrated with Amazon Deequ for automated data quality audits across diverse data ingestion and transformation stages. - Automated end-to-end workflow execution, programmatic S3 report management, and instant Slack notifications for operational status tracking using Python and Airflow. - Created functionality to automatically generate comprehensive JSON and HTML profiling reports for streamlined stakeholder reviews.

AWARDS & HONORS

Gold Medalist – MCA (Software Engineering) – Awarded the University Gold Medal for securing the first rank across the 2023–2025 batch.
Student of the Year 2023 (PGDAV College, Delhi University) – Honored for outstanding academic performance, leadership, and proactive technical contributions across college societies.